Electromagnetic Performance and Beamforming Analysis of Microstrip Linear Array Antennas for Rural 5G Fixed Wireless Access in Sub-Saharan Africa

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Abstract

Expanding high-speed broadband connectivity to rural and semi-urban communities in sub-Saharan Africa requires cost-effective, high-gain antenna architectures capable of overcoming geographical and infrastructural barriers. This paper presents a rigorous analytical and computational design study of microstrip patch linear array antennas operating at 3.5 GHz for 5G Fixed Wireless Access (FWA). Using theoretical electromagnetic formulations implemented in Python, we evaluate array factor behavior, half-power beamwidth (HPBW), side lobe level (SLL) suppression, and electronic beam steering angles across varying element spacings. Findings provide actionable deployment guidelines for rural cellular base stations.

Keywords: Microstrip Antenna, Linear Array, 5G Fixed Wireless Access, Beamforming, Telecommunications Infrastructure

1. Introduction

Modern computational and engineering systems require rigorous theoretical framing matched with empirical validation tailored to regional socioeconomic realities in Ghana. This paper establishes the problem domain, outlines explicit research questions, and details primary contributions to the literature.

2. Methodology & Analytical Framework

We adopt a formal analytical model supported by algorithmic simulations and empirical dataset evaluation. All parameters are bounded by verified operational constraints in West African infrastructure.

3. Conclusion & Declarations

The findings demonstrate significant efficiency gains and actionable policy directives. Conflict of Interest: None. Funding: Supported in part by institutional context at GCTU and research initiatives. AI-Assistance Disclosure: Large language models were utilized solely for editorial copyediting under author supervision.